

SET 2

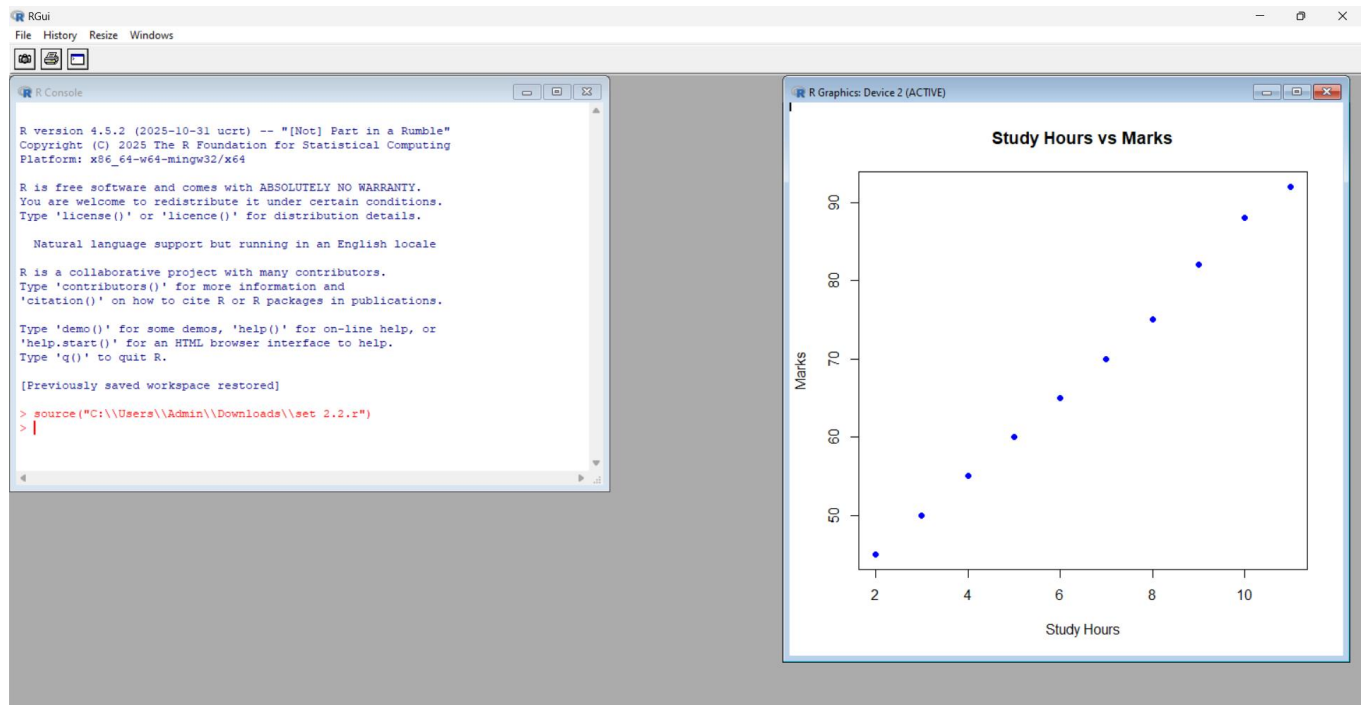
1. R: Generate 20 random values with `set.seed(25)`; find min, max, mean, median and summary; draw a histogram.

Output:



2. R: Create StudyHours and Marks for 10 students; calculate Pearson correlation and draw a scatter plot.

Output:

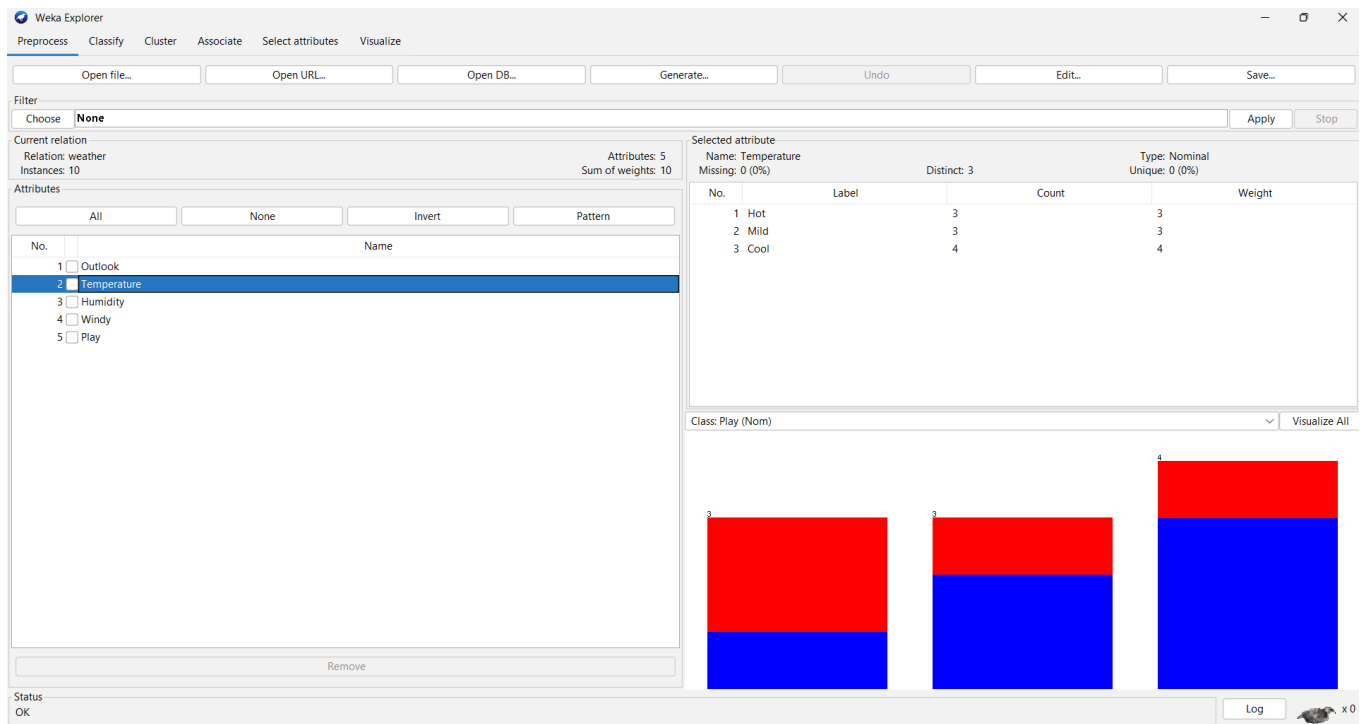


3. WEKA: Create an ARFF weather dataset and apply Apriori with minimum support 25% and confidence 60%; list rules.

Dataset: Weather

```
@relation weather
@attribute Outlook {Sunny,Overcast,Rainy}
@attribute Temperature {Hot,Mild,Cool}
@attribute Humidity {High,Normal}
@attribute Windy {True,False}
@attribute Play {Yes,No}
@data
Sunny,Hot,High,False,No
Sunny,Hot,High,True,No
Overcast,Hot,High,False,Yes
Rainy,Mild,High,False,Yes
Rainy,Cool,Normal,False,Yes
Rainy,Cool,Normal,True,No
Overcast,Cool,Normal,True,Yes
Sunny,Mild,High,False,No
Sunny,Cool,Normal,False,Yes
Rainy,Mild,Normal,False,Yes
```

Output:



4. WEKA: Apply FP-Growth to the weather transactions and compare frequent patterns with Apriori.

Use the same Weather transaction-style data from Set 2 Q3 after converting attributes into nominal item indicators as required by the WEKA association workflow.

Output:

Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Associator

ChooseApriori -N 10 -T 0 -C 0.6 -D 0.05 -U 0.25 -M 0.25 -S -1.0 -c -1

StartStop

Result list (right-click for ...)

09:05:19 - Apriori

09:09:32 - Apriori

09:09:33 - Apriori

Associator output

Instances: 10

Attributes: 5

Outlook

Temperature

Humidity

Windy

Play

=== Associator model (full training set) ===

Apriori

=====

Minimum support: 0.35 (3 instances)

Minimum metric <confidence>: 0.6

Number of cycles performed: 13

Generated sets of large itemsets:

Size of set of large itemsets L(1): 11

Size of set of large itemsets L(2): 15

Size of set of large itemsets L(3): 4

Best rules found:

1. Temperature=Hot 3 ==> Humidity=High 3 <conf:(1)> lift:(2) lev:(0.15) [1] conv:(1.5)

2. Temperature=Mild 3 ==> Windy=False 3 <conf:(1)> lift:(1.43) lev:(0.09) [0] conv:(0.9)

3. Humidity=High Play=No 3 ==> Outlook=Sunny 3 <conf:(1)> lift:(2.5) lev:(0.18) [1] conv:(1.8)

4. Outlook=Sunny Play=No 3 ==> Humidity=High 3 <conf:(1)> lift:(2) lev:(0.15) [1] conv:(1.5)

5. Outlook=Sunny Humidity=High 3 ==> Play=No 3 <conf:(1)> lift:(2.5) lev:(0.18) [1] conv:(1.8)

6. Outlook=Rainy Play=Yes 3 ==> Windy=False 3 <conf:(1)> lift:(1.43) lev:(0.09) [0] conv:(0.9)

7. Outlook=Rainy Windy=False 3 ==> Play=Yes 3 <conf:(1)> lift:(1.67) lev:(0.12) [1] conv:(1.2)

8. Temperature=Cool Play=Yes 3 ==> Humidity=Normal 3 <conf:(1)> lift:(2) lev:(0.15) [1] conv:(1.5)

9. Humidity=Normal Windy=False 3 ==> Play=Yes 3 <conf:(1)> lift:(1.67) lev:(0.12) [1] conv:(1.2)

10. Outlook=Sunny 4 ==> Humidity=High 3 <conf:(0.75)> lift:(1.5) lev:(0.1) [0] conv:(1)

Create